

→ logical equivalence

$$P \Rightarrow Q = \sim P \vee Q$$

P	Q	H.C.	$P \Rightarrow Q$	$\sim P \vee Q$	$P \Leftrightarrow Q$
T	T		T	T	T
T	F		F	F	F
F	T		T	T	F
F	F		T	T	T

$P \Leftrightarrow Q$ (highlighted in orange)

$P \Rightarrow Q \wedge Q \Rightarrow P$ (circled in green)

$(P \wedge Q) \wedge (\sim P \wedge \sim Q)$ (circled in green)

$(\sim P \vee Q) \wedge (\sim Q \vee P)$ (circled in green)

$\forall \rightarrow$ uni.

$\exists \rightarrow$ exs.

Goldbach Conjecture:

$P \rightarrow$ set of primes

$X \rightarrow$ set even no. > 2

$$(n \in X) \rightarrow (\exists q, p \in P, n = p + q)$$